

Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer

ABSTRACT

Air pollution especially fine particulate matter PM2.5 can present several risks to not only human cardiovascular health, but it can also affect the neurological health as well. This study focuses on the relationship between PM2.5 exposure and the cognitive and motor performance of participants who completed an online motor learning task called Super G. For the Super G task 1,397 participants across the United States provided their zip codes and completed the motor task which measured their learning and performance. Air Quality data from 2022 to 2024 were collected from the Environment Protection Agency (EPA) and geocoded to the Participant zip code to the nearest monitoring sites. The result showed that participant with higher exposure to PM2.5 demonstrated little to no improvement whereas low PM2.5 exposed participant showed greater improvement. Through these finding, it can be implied that long term chronic exposure to PM2.5 can cause neurocognitive impairment which can potentially lead to neurodegenerative diseases such as Parkinson's, Alzheimer's and Dementia.

KEYWORDS: PM2.5, air pollution, fine particulate matter, cognitive, motor, neuroinflammation, oxidative stress, neurocognitive function, brain health, EPA data, geocoding, motor learning, central nervous system, long-term exposure, neurodegeneration, Alzheimer's disease, Parkinson's disease, dementia, air quality index (AQI), environmental neuroscience,

I. INTRODUCTION

Air quality plays an important role in both environmental and human health. A wide range of health complications have risen from exposure to polluted air.¹ Such health complications include respiratory, and neurological disorders.¹ As the world progresses to urbanization, industrialization and increase in usage of fossil fuels, pollution related health risks are increasing in prevalence and severity. Air pollution contain toxic and harmful substances such as Particulate Matter (PM), Carbon Monoxide (CO), Ozone, Nitrogen dioxide (NO₂), sulfur dioxide (SO₂) and other pollutants. Among particulate matter, PM2.5 which are particles less than 2.5 micrometer diameter length are known to be an environmental health hazard. It can penetrate the respiratory system and enter the bloodstream.

Vehicle emissions, factories, burning of fossil fuels are the primary sources of PM2.5 released into the air. PM2.5 can infiltrate the various organs including the brain.² PM2.5 enters the body when it is inhaled and goes through the body's respiratory system, lungs and crosses toward the circulatory system and eventually enters the blood stream. A study has shown the PM2.5 can enter the central nervous system by systemic circulation or olfactory pathway. When PM2.5 reaches the brain, a cascade of inflammatory response is triggered.³ These inflammatory responses can cause an activation of microglial, hence may cause the microglia to enter an overactivated state and cause neurodegenerative disorders.⁴ PM2.5 can also cause oxidative stress⁴ and neurovascular dysfunction.⁵ Since PM2.5 contain oxidants, it can induce oxidative stress, increasing the effect of neurodegeneration due to the aggravation of glutamatergic excitotoxicity in neurons.² Air pollution does not only affect the brain via microglial cells. Studies have shown that astroglia cells are also affected by PM2.5.⁶ PM2.5 hyperactivates the astrocytes and increased its reactivity causing a rise in oxidative stress and inflammation in the brain.⁶ Inflammation can also occur in the blood brain barrier.⁵ A study has shown that when the PM2.5 is crosses the blood brain barrier, PM2.5 accumulates in the brain tissue which can initiate an inflammation response.⁶ The negative effect of

neurodegeneration can cause risk of diseases such as Alzheimer's disease, Parkinson's and Dementia.⁵ PM2.5 can affect cognition and motor through memory problems, and attention deficits.⁷ Studies have shown that selective attention and emotion expression have reduced after exposure to PM air pollution which links to previous research regarding inflammation.⁷ PM2.5 have also shown to affect the body physically which adds up to the delayed response. Long term PM2.5 exposure can also increase the chances of Sarcopenia risk which indicates a gradual loss of muscle mass and strength.⁸

Air quality consists of a variety of different components which are PM2.5, PM10, Ozone, Carbon Monoxide, Nitrogen Dioxide, Sulfur Dioxide and lead. PM2.5 is the most toxic component listed which is why the objective of this experiment mostly focuses on that particle. It has the ability to not only penetrate deep into the lungs and the blood stream, but even small amounts of this component can cause heart disease, stroke, and other respiratory and neurological disorders. The negative effect of this air quality component can attack multiple organs instead of just one. Although there is strong evidence for the biological impact of PM2.5 on the central nervous system, and exposure to PM2.5 may lead to reduced memory and attention, it has to be demonstrated if exposure to PM2.5 leads to reduction in motor skill.

The objective is to analyze the correlation between poor air quality and motor performance. This link will allow us to understand how air pollution, specifically PM2.5 impacts human behavior, specifically the ability to learn a motor skill. Understanding this will help us understand the harmful impact of poor air quality to humans.

II. MATERIALS AND METHODS

Through the Amazon Mechanical Turk (MTurk) platform, a total of 1397 participants were recruited. These participants have provided an informed consent before taking part in this experiment. The Institution Review Board of Arizona State University has also approved this study

Shivangi Sarkar

Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer

((Study 00015247).

MOTOR SKILL TEST

The participants performed a test where their motor skill was assessed across a two-year time period. Participants play an online motor task called “Super G” (a physics-based game) where the participants control an astronaut. The goal for the participants is to land the astronaut, using either their computer arrow keys or touch screen, to as many planets as possible.⁹ After they have completed 75 trials, there performance metrics were recorded. The metrics included astronaut’s position and acceleration, the date and time of each trial, the number of trials completed, the highest score, the total planets landed on, response time and time in target.⁷ Along with the performance metrics, the participants completed a survey which asked them to provide their 5-digit zip code. Based on their zip code, environmental and socioeconomic variables such as Area Deprivation Index, Water Quality, Air Quality, and Food Access were derived. This analysis will focus on air quality only.

AIR QUALITY AND SOCIOECONOMIC DATA COLLECTION

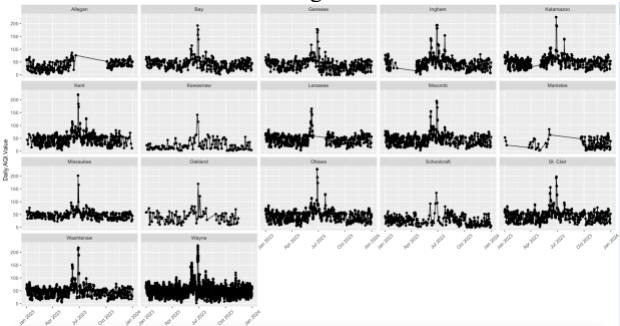
Air quality data was accessed by the Environmental Protection Agency for the year 2022, 2023 and 2024. Data analysis and visualization was performed in R. Mean AQI and PM2.5 were computed for each county near a monitoring site. Participant zip codes were geocoded to the nearest air quality monitoring sites using geosphere package. The AQI and PM2.5 data from the nearest monitoring sites were assigned to the participants data. Visualization was then performed using ggplot package to represent the linear trends between Super G motor performance and air quality.

III. RESULTS

AIR QUALITY INDEX (AQI) IN 2023 AND 2024

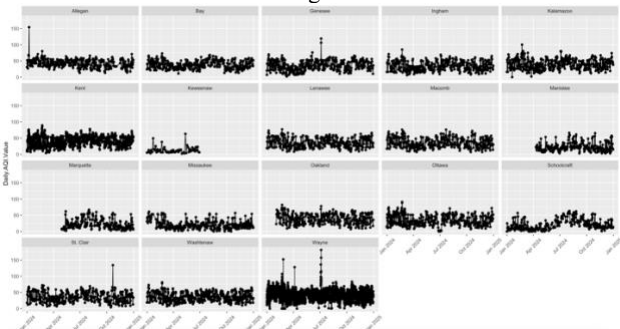
Air Quality data from the environmental protection agency (EPA) were collected to analyze the daily AQI values across the different counties in USA. Figure 1 and 2 shows the monthly AQI values of 2023 and 2024 respectively and Figures 3 and 4 shown illustrates how AQI values are distributed across different counties respectively. The higher bars represent AQI values being more prevalent. Since most counties have AQI values predominantly being below 50, it indicates good air quality however there are some counties with elevations in AQI which suggests air pollution being present.

Figure 1



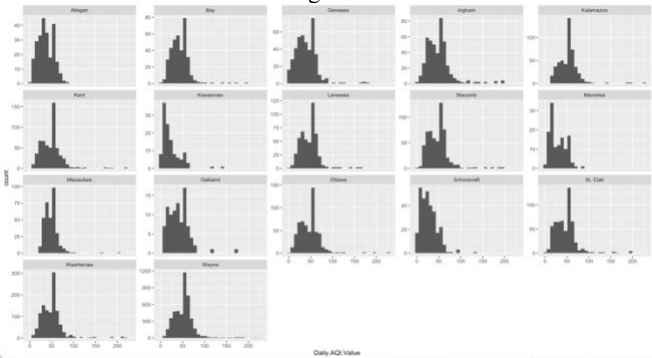
2023 Air quality data in different counties of the US.

Figure 2



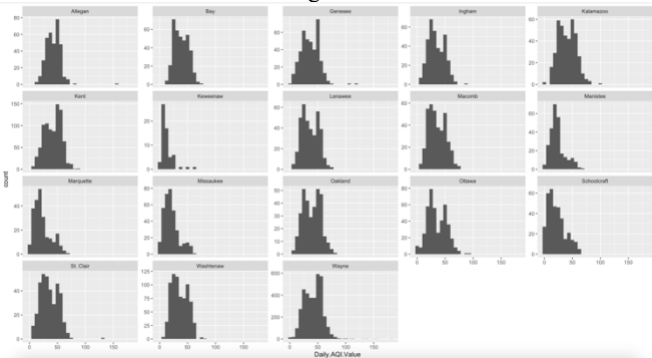
2024 Air quality data in different counties of the US.

Figure 3



2023 data of Common AQI values in different counties.

Figure 4



2024 data of Common AQI values in different counties.

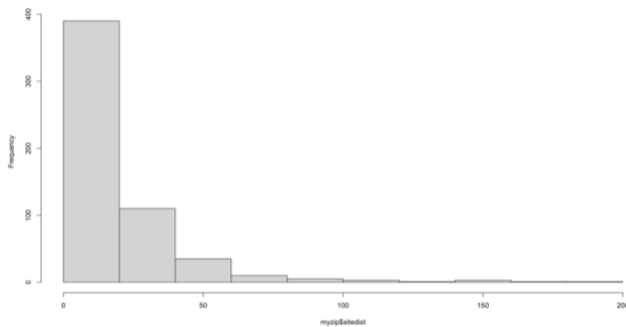
Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer

PROXIMITY OF AIR QUALITY MONITORING SITES TO THE PARTICIPANTS

Participants' locations were analyzed in relation to the nearest air quality monitoring sites. Figure 5 (2023) and figure 6 (2024) shows the distance distribution between the zip codes of the participants and the nearest AQI monitoring sites. Majority of the participants live within 10 km of the monitoring site which represents accurate allocation. The participants located more than 50 km away from the nearest cite may be contribute to uncertainty in air quality exposure.

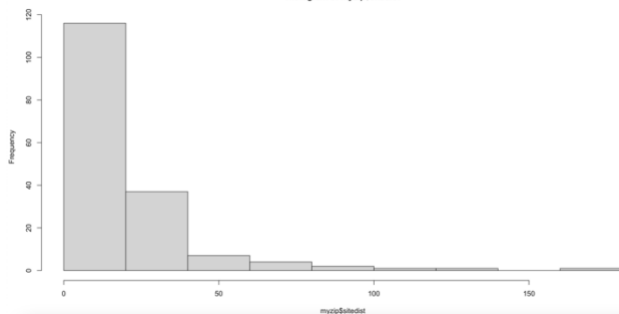
Figure 5



2023 Participant data of distance from participant zip code to its nearest monitoring sites

Figure 6

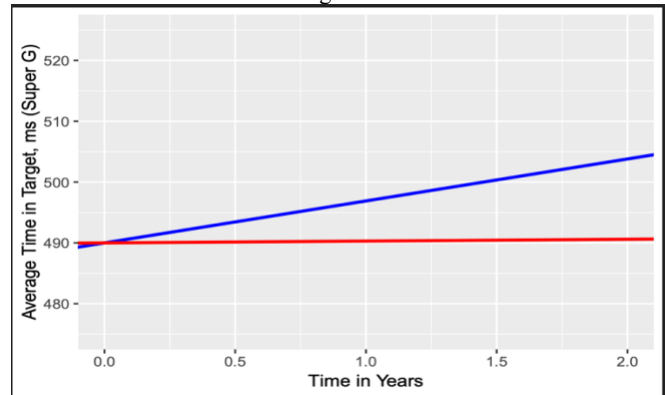
Histogram of mzipdist



2024 Participant data of distance from participant zip code to its nearest monitoring sites

PM 2.5 EFFECTS ON COGNITIVE-MOTOR PERFORMANCE
Player's performance on the Super G task was analyzed with respect to PM2.5 exposure. Participants were separated into high and low exposure groups to PM2.5. In Figure 7, the graph depicts participants exposed to lower PM2.5 levels (blue line) shown greater improvement of the task whereas the participants exposed to higher PM2.5 level (red line) demonstrates a constant line of no improvement suggesting detrimental effect of air pollution on neurocognitive function.

Figure 7

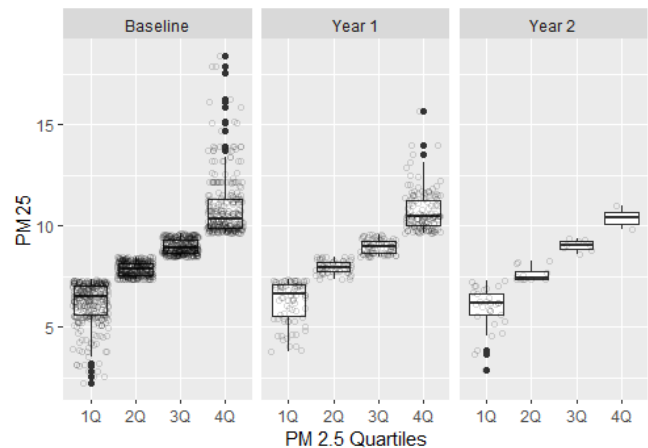


High PM2.5 (red) and low PM2.5(blue) exposed participants improvement of the task over years

PM 2.5 EXPOSURE ANALYSIS AND MOTOR PERFORMANCE TRENDS

Participants data were categorized into quartile based on their exposure to PM2.5. 1Q, 2Q, 3Q, and 4Q is low exposure, low-medium exposure, medium exposure, and high exposure respectively. In the Box Plot it was shown that as time progressed from baseline to year 2, fewer participants are exposed to a lot of pollution.

Figure 8



Participant's exposure to air quality over baseline to Year 2

IV. DISCUSSION

Based on the results we could see a correlation between higher PM 2.5 exposure and impaired cognitive and motor performance over time. Participants with lower PM 2.5 exposure levels have shown improvements on the motor task while higher levels have shown no improvement. The results have hence shown how air pollution hinder neurocognitive learning process.

PM 2.5 have shown to trigger harmful processes in the central nervous system. PM2.5 crosses the first barrier

Shivangi Sarkar

Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer

which is the pulmonary barrier and then enters system circulation. PM 2.5 eventually enters the blood brain barrier via direct transport or systemic inflammation.¹⁰ PM 2.5 in the brain creates microglial activation which leads to neuroinflammation. Microglia overactivation can release of pro inflammatory cytokines and reactive oxygen species which can increase oxidative stress and eventually cause neuronal damage.¹¹

Inflammation also activates astrocytes which amplifies oxidative stress and disrupts neuronal functions.¹² PM2.5 can also compromise the integrity of the Blood Brain Barrier which increases its permeability and causes other foreign particles to enter the brain.¹³ Eventually these disruptions validate the results of decrease participant performance which is occurring due to the impairments in motor and cognitive function in the high exposure group.

Super G task tests the cognitive-motor skills by require the participant ability to focus, remember and coordinate fluently. The brain's ability to learn and remember mostly depends on how much neuroinflammation and oxidative stress have impaired the synaptic plasticity and the Long Term potentiation.¹⁴ Since the figures have shown a decrease in improvement of motor task performance associated with high PM2.5 exposed participants; air pollution does negatively affect neural process. PM2.5 not only affects the brain, but it can also cause muscle atrophy and Sarcopenia which can play part in poor motor performance.¹⁵

These findings supported by these figures indicate how long-term exposure to PM2.5 can increase risk of neurodegenerative diseases such as Alzheimer's, Parkinson's, and Dementia.^{10,11} Due to cognitive and motor deficits caused by neurological impairments, neurodegenerative diseases are emerging. In Parkinson's, the increase oxidative stress and neuroinflammation induced by PM2.5 affects dopaminergic neurons in Substantia Nigra.¹¹ Substantia nigra is critical region for motor control. Any negative effect in substantia nigra like loss of neurons in the area can cause tremors, and brady pollutants like PM2.5 can induce α -synuclein aggregation. A-synuclein can disrupt proteostasis and mitochondrial health which also plays a part in motor function. In Alzheimer's, accumulation of amyloid beta plaques and tau protein disrupts synaptic communication and can cause neuronal death.¹⁴ Microglial activation caused by PM2.5 exposure have shown to increase amyloid plaque accumulation and neurodegeneration. Although the findings pose substantial evidence to negative impact of cognitive-motor performance affected by poor air quality, there were limitation to the study. One of the limitations were the distance between air monitoring stations and the participant location. This may have caused a slightly inaccurate air quality Index and PM2.5 exposure value. Another limitation may have been other pollutants such as NO2 and Ozone affecting the AQI index and PM2.5 value. Another possible sources of error may have the Participant sample size decreasing over time in the experiment.

V. CONCLUSION

This study shows evidence that exposure to high levels of PM2.5 can negative affect cognitive motor performance over time. Participant with high exposure to PM2.5 shown poor scores in the Super G task compared to those with lower exposure due to harmful effects of PM 2.5. Prolonged exposure to poor air quality can lead to risk of various different neurodegenerative diseases such as dementia, Alzheimer's, and Parkinson's.

Even though limitations are present in this study, the results do show that there is an urgent need for air quality regulation which aims to reduce the fine particulate matter. It can protect population from cardiovascular risk as well neurological degeneration form occurring.

Future studies could include in vivo animal model which will observed to understand the impact of PM2.5 on brain structure, motor function and inflammation. Effects of poor air quality could also be analyzed in areas where the air quality index is terrible such as China, and India.

VI. REFERENCES

1. Liu, C., Chen, R., Sera, F., Vicedo-Cabrera, A. M., Guo, Y., Tong, S., ... & Gasparri, A. (2020). Ambient particulate air pollution and daily mortality in 652 cities. *The New England Journal of Medicine*, 381(8), 705-715. <https://doi.org/10.1056/NEJMoa1817364>
2. Zhang, X., Chen, X., Zhang, X., Mao, Z., Jiang, S., Wang, H., & Zhang, Y. (2022). The effects of PM2.5 exposure on brain: A systematic review and meta-analysis. *Journal of Biomedical Science*, 29, 79. <https://doi.org/10.1186/s12929-022-00799-x>
3. Meo, S. A., Shaikh, N., Alotaibi, M., AlWabel, A. A., & Alqumaidi, H. (2024). Effect of air pollutants particulate matter (PM_{2.5}, PM₁₀), sulfur dioxide (SO₂) and ozone (O₃) on cognitive health. *Scientific Reports*, 14, Article 19616. <https://doi.org/10.1038/s41598-024-70646-6>
4. Song, J., Han, K., Wang, Y., Qu, R., Liu, Y., Wang, S., Wang, Y., An, Z., Li, J., Wu, H., & Wu, W. (2022). Microglial activation and oxidative stress in PM_{2.5}-induced neurodegenerative disorders. *Antioxidants*, 11(8), 1482. <https://doi.org/10.3390/antiox11081482>
5. You, R., Ho, Y.-S., & Chang, R. C.-C. (2022). The pathogenic effects of particulate matter on neurodegeneration: A review. *Journal of Biomedical Science*, 29, Article 15. <https://doi.org/10.1186/s12929-022-00799-x>
6. Kang, Y. J., Tan, H. Y., Lee, C. Y., & Cho, H. (2021). An air particulate pollutant induces neuroinflammation and neurodegeneration in human brain models. *Advanced Science*, 8(21), 2101251. <https://doi.org/10.1002/advs.202101251>
7. Sukumaran, K., Botterhorn, K. L., Schwartz, J., Gauderman, J., Cardenas-Iniguez, C., McConnell, R., Hackman, D. A., Berhane, K., Ahmadi, H., Abad, S., Habre, R., & Herting, M. M. (2024, October 30). Associations between fine particulate matter components, their sources, and cognitive outcomes in children ages 9–10 years old from the United States. *Environmental Health Perspectives*, 132(10), Article 107009. <https://doi.org/10.1289/EHP14418>
8. Chen, S.-W., Lin, C.-Y., Chen, C.-Y., Lin, C.-L., Hsieh, T.-L., Tsai, F.-J., & Chang, K.-H. (2023). Long-term exposure to air pollution and risk of sarcopenia in adult residents of Taiwan: A nationwide retrospective cohort study. *BMC Public Health*, 23, Article 2172. <https://doi.org/10.1186/s12889-023-17091-8>
9. Hooyman, A., & Schaefer, S. Y. (2023). Age and sex effects on

Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer

- Super G performance are consistent across internet devices. *International Journal of Serious Games*, 10(2), 25–36. <https://doi.org/10.17083/ijsg.v10i2.598>
10. Zhang, X., Chen, X., Zhang, X., Mao, Z., Jiang, S., Wang, H., & Zhang, Y. (2022). The effects of PM_{2.5} exposure on brain: A systematic review and meta-analysis. *Journal of Biomedical Science*, 29, 79. <https://doi.org/10.1186/s12929-022-00799-x>
 11. Song, J., Han, K., Wang, Y., Qu, R., Liu, Y., Wang, S., Wang, Y., An, Z., Li, J., Wu, H., & Wu, W. (2022). Microglial activation and oxidative stress in PM_{2.5}-induced neurodegenerative disorders. *Antioxidants*, 11(8), 1482. <https://doi.org/10.3390/antiox11081482>
 12. You, R., Ho, Y.-S., & Chang, R. C.-C. (2022). The pathogenic effects of particulate matter on neurodegeneration: A review. *Journal of Biomedical Science*, 29, Article 15. <https://doi.org/10.1186/s12929-022-00799-x>
 13. Kang, Y. J., Tan, H. Y., Lee, C. Y., & Cho, H. (2021). An air particulate pollutant induces neuroinflammation and neurodegeneration in human brain models. *Advanced Science*, 8(21), 2101251. <https://doi.org/10.1002/advs.202101251>
 14. Meo, S. A., Shaikh, N., Alotaibi, M., AlWabel, A. A., & Alqumaidi, H. (2024). Effect of air pollutants particulate matter (PM_{2.5}, PM₁₀), sulfur dioxide (SO₂) and ozone (O₃) on cognitive health. *Scientific Reports*, 14, Article 19616. <https://doi.org/10.1038/s41598-024-70646-6>
 15. Chen, S.-W., Lin, C.-Y., Chen, C.-Y., Lin, C.-L., Hsieh, T.-L., Tsai, F.-J., & Chang, K.-H. (2023). Long-term exposure to air pollution and risk of sarcopenia in adult residents of Taiwan: A nationwide retrospective cohort study. *BMC Public Health*, 23, Article 2172. <https://doi.org/10.1186/s12889-023-17091-8>

Effects of Air Quality on Cognitive and Motor Scores

Shivangi Sarkar, Andrew Hooyman, Sydney Schaefer